

Travel Tax: NBA Schedule Friction & Competitive Equity

Five-season geospatial analysis by Samir Sadman

Dataset: 6,000 NBA regular-season games / 12,000 team-game observations, 2020-21 through 2024-25.

Primary result: +1 unit of relative Schedule Friction Index is associated with a -1.35-point change in home-team margin (95% CI -2.06 to -0.63; $p=0.0002348$).

Win-probability model: average -4.0 percentage-point change in predicted home-win probability per +1 relative-SFI unit.

Schedule Friction Index components

Log travel distance; absolute time-zone shift; rest deficit; games in the previous seven days; consecutive road-game streak; positive altitude gain. Components are standardized within season and outcomes are not used to construct the index.

Primary specification

home margin $\sim$ relative SFI + pre-game 10-game form differential + season fixed effects + home-team fixed effects + away-team fixed effects; HC3 robust standard errors.

Interpretation

The results are observational. They are consistent with schedule geography and recovery burden having a measurable association with game performance, but they are not a causal estimate of fatigue.